PERSONAL INFORMATION

Family name, First Name	Iribarren Aguirre, Iñigo
E-mail	inigo.iribarren@tum.de
Bluesky handle	@iiribarren.bsky.social
GitHub	github.com/iribirii
ORCID	0000-0003-0373-8687

PROFESIONAL EXPERIENCE

Mar 2024- present	**Post-doctoral researcher** in Computational Bioinorganic Medicinal Chemistry, School of Natural Sciences, Technical University of Munich, Germany. *Machine Learning-Powered Design of Supramolecular Metallacages as Novel Drug Delivery Systems*
Sep 2019- Jan 2024	**Ph.D.** in Computational Organic Chemistry, School of Chemistry, Trinity College Dublin (TCD), Ireland. *Theoretical development of new class of phase transfer catalysts: applications in the pharmaceutical industry*
Nov 2018 - Aug 2019	**Consultant** at Everis (NTT Data), Madrid, Spain Big Data projects at Banco Santander and Vodafone
Sep 2016 - Jul 2018	**M.Sc.** in Theoretical Chemistry and Computational Modelling (TCCM), Universidad Autonoma de Madrid (UAM), Madrid, Spain. *Cations brought together by hydrogen bonds: The protonated pyridine–boronic acid dimer explained*

FELLOWSHIPS & AWARDS

2026	**Best Contributed Talk** - European Symposium on Chemical Bonding (CBOND2026) *A Dynamic Evaluation of Cisplatin Encapsulation into* $[\mathrm{Pd_2L_4}]^{4+}$ *Metallacages in Solution*
2024	BOC Gases bursary thesis competition - 2nd prize
2022	**Best poster** - 12th Triennial Congress of the World Association of Theoretical and Computational Chemists (WATOC 2020) *Efficiency and Suitability when Exploring the Conformational Space of Phase-Transfer Catalysts*
2022	**Best poster** - 3rd European Symposium on Chemical Bonding (CBOND2022) *Efficiency and Suitability when Exploring the Conformational Space of Phase-Transfer Catalysts*
2020	**Best flash talk** - Institue of Chemistry of Ireland Postgraduate Chemistry Research Symposium 2020 *Anion Recognition by Neutral and Cationic Iodotriazole Halogen Bonding Scaffolds*
2019-2023	SFI fellowship for Ph.D. studies
2016-2018	UAM-Fujitsu fellowship for TCCM master studies

RESEARCH VISITS

May-July 2022	Medicinal Chemistry Institue (IQM-CSIC), Madrid, Spain *Prof. Ibon Alkorta*
May 2022	École Polytechnique Fédérale de Lausanne (EPFL), Laussane, Switzerland *Prof. Clémence Corminboeuf*

SUPERVISION OF STUDENTS

2026	Technical University of Munich **Master internship student** Tony Benkert - *UV-VIS absortion spectra prediction for Au-based porphyrins*
2021-2022	Trinity College Dublin **Senior Sohister final project** Diana Bura - *Theoretical Study of Phase-Transfer Catalysts for asymmetric conjugate cyanation of α-β-unsaturated ketones*
2021-2022	Trinity College Dublin **Senior Sophister final project** John Martyn - *Theoretical Study of Phase-Transfer Catalysts for asymmetric conjugate cyanation of α-β-unsaturated ketones*
2021	Trinity College Dublin **Summer project** Diana Bura - *Conformational Study of Phase-Transfer Catalysts*
2020-2021	Technological University Dublin **Senior Sophister final project** Nika Melnyk - *Docking vs Quantum Mechanical calculations for the Analysis of Anti-malarial 1,2,4,5-tetraoxanes - based compounds*
2020-2021	Trinity College Dublin **Senior Sophister final project** Harris Shariff - *Study of Tuneable Bioactive Anion Transporters*
2020	Trinity College Dublin **Summer project** Sean O'Conell - *Conformational Study of Phase-Transfer Catalysts*
2019-2020	Technological University Dublin **Senior Sophister final project** Cathal Barry - *Computational analysis of antimalarial 1,2,4-trioxolane and 1,2,4,5-tetraoxanes based compounds*

TEACHING

2025	Python for Engineering Data Analysis - **Assistant lecturer**
2025	Python for Engineering Data Analysis - **Assistant lecturer**
2023	CHU33603, CHU33309, CHU33509 - Practical in Physical chemistry - Hückel lab - **Senior Demonstrator**
2022	CHU33603, CHU33309 - Practical in Physical chemistry - Hückel lab - Regular Demonstrator
2022	CHU33409 - Analytical and computational methods - Interdisciplinary Computational Organic Chemistry lab - Regular Demonstrator
2021	TUD Computational Methods in Organic Chemistry practicals - **Senior Demonstrator**
2021	CHU33409 - Analytical and computational methods - Interdisciplinary Computational Organic Chemistry lab - Regular Demonstrator
2021	CHU33442 - Drug design and development - QSAR lab - Regular Demonstrator
2020	CHU33442 - Drug design and development - QSAR lab - Regular Demonstrator
2020	CHU33409 - Analytical and computational methods - Interdisciplinary Computational Organic Chemistry lab - Regular Demonstrator
2019	CHU11B01 - Chemistry for Biologists - Computational lab - Regular Demonstrator

MEMBER OF SCIENTIFIC SOCIETIES

Since 2029	Membership of Royal Spanish Society of Chemistry

ORGANISATION OF SCIENTIFIC MEETINGS

2026 Cheminformatics, Automation and Machine Learning in Chemistry micro-credential
Organising committee

2025 Cheminformatics, Automation and Machine Learning in Chemistry micro-credential
Organising committee

2024 Cheminformatics, Automation and Machine Learning in Chemistry micro-credential
Organising committee

2023 ChemEd-Ireland Conference 2023, Tinity College Dublin, Dublin, Ireland.
Organising committee

Invited Seminars

2026 Technical University of Munich - AMC Seminar Afternoon

2025 Vrije Universiteit Brussels - Chemical Theory Group

CONFERENCES

2026 European Symposium on Chemical Bonding (CBOND2026) - *Talk* - *poster*
A Dynamic Evaluation of Cisplatin Encapsulation into $[Pd_2L_4]^{4+}$ *Metallacages in Solution*

2026 17th World Congress on Computational Mechanics - 10th European Congress on Computational Methods in Applied Sciences and Engineering (WCCM-ECCOMAS2026) - *Talk*
A Computational Workflow to Unravel the Structural Dynamic of Supramolecular Metallacages in Solution

2026 Chemical Compound Space Conference (CCSC2026) - *poster*
A Computational Workflow to Unravel the Structural Dynamic of Supramolecular Metallacages in Solution

2025 Theoretical Chemistry and Computational Modeling: 25 years promoting Excellence in Science (25TCCM) - *oral communication*
Computational workflow comparison to study new anti-cancer drug delivery systems

2025 The 13th OpenMolcas Developers' Workshop

2025 Global Young Scientists Summit (GYSS2025) - *poster*
Machine Learning-Powered Design of Supramolecular Metallacages as Drug Delivery Systems

2024 CeNS-e-conversion workshop at Venice International University - *poster*
Machine Learning-Powered Design of Supramolecular Metallacages as Drug Delivery Systems

2024 MOF Computational Tools and Machine Learning

2024 4th MDSI General Assembly - *poster*
Machine Learning-Powered Design of Supramolecular Metallacages as Drug Delivery Systems

2024 1st Annual Meeting EIC Pathfinder Open "SMARTdrugs"

2023 European Conference on Computational & Theoretical Chemistry (EuChemS 2023) - *poster*
Simultaneous Hydrogen Bonds with Different Binding Modes: The Acceptor "Rules" but the Donor "Chooses"

2023 Royal Society of Chemistry annual Organic Chemistry in Ireland - *poster* (July)
Simultaneous Hydrogen Bonds with Different Binding Modes: The Acceptor "Rules" but the Donor "Chooses"

2023 XXXIX Bienal de la Real Sociedad Española de Química (BQZ2023) - *flash communication*
Simultaneous Hydrogen Bonds with Different Binding Modes: The Acceptor "Rules" but the Donor "Chooses"

2023 3 minutes thesis competition - *oral communication*
Evolution of design approaches in asymmetric organocatalysis over the last decade

2023 Tetrahedron Chem Seminar Series - *oral communication*
Theoretical development of new class of phase transfer catalysts: applications in the pharmaceutical industry

2023 The Conference of Young Researchers in Atomic and Molecular Physics (J2IFAM) - *oral communication*
Exploring the conformational space within phase transfer catalysts: computational methods benchmarking

2023 Young Scientists Workshop on Theoretical Chemistry and Computational Modelling (YS TCCM 2023) - *oral communication*
Theoretical development of new class of phase transfer catalysts: applications in the pharmaceutical industry

2022 12th Triennial Congress of the World Association of Theoretical and Computational Chemists (WATOC 2020) - *poster* - **best poster award**
Exploring the conformational space within phase transfer catalysts: computational methods benchmarking

2022 3rd European Symposium on Chemical Bonding (CBOND2022) - *poster* - **best poster award**
Exploring the conformational space within phase transfer catalysts: computational methods benchmarking

2022 12th Congress on Electronic Structure Principles and Applications (ESPA-2022) - *poster*
Exploring the conformational space within phase transfer catalysts: computational methods benchmarking

2022 LatinXChem2023 - *poster*
Exploring the conformational space within phase transfer catalysts: computational methods benchmarking

2021 Virtual GEQC-RSEQ Symposium - *flash talk*
Evaluation of Electron Density Shifts in Noncovalent Interactions

2021 European Symposium on Organic Reactivity (ESOR2021) - *poster*
Improving phase-transfer catalysis by enhancing non-covalent interactions

2021 LatinXChem2021 - *poster*
Evaluation of Electron Density Shifts in Noncovalent Interactions

2021 Royal Society of Chemistry twitter poster event *poster*
Anion Recognition by Neutral and Cationic Iodotriazole Halogen Bonding Scaffolds

2020 Institue of Chemistry of Ireland Postgraduate Chemistry Research Symposium 2020 - *flash talk* - **best talk award**
Anion Recognition by Neutral and Cationic Iodotriazole Halogen Bonding Scaffolds

2020 Virtual Conference on Theoretical Chemistry - *poster*
Anion Recognition by Neutral and Cationic Iodotriazole Halogen Bonding Scaffolds

OUTREACH AND SCIENCE COMMUNICATION

2026 SciLab Deutsches Museum - 150th Electrical engineering and information technology at TUM anniversary - **Activity leader**

2023 ESB Science Blast 2023 - **Activity leader**

2021-2024 Current Chemistry Investigators - Secondary schools workshops - **Activity leader** *and ambassador*

2020-2021	Stories from the lab - Secondary schools workshops - *Activity ambassador*
2020	European researchers night - Soap vs COVID-19 website - *Presenter*
2019-2020	Spectroscopy in a suitcase - Secondary schools workshops - *Activity ambassador*

RESEARCH ACTIVITY

ORCID-ID	0000-0003-0373-8687
Total peer-reviewed publications	19
h-index	8 Scopus

[19] I. Iribarren, T. Renningtlotz, and C. Trujillo. "In silico Catalyst Design Within Asymmetric Organocatalysis". In: *Enantioselective Organocatalysis: Catalysts, Reactions, and Applications* 1 (2026), pp. 225–261.

[18] G. Moreno-Alcántar, C. Schmidt, D. D. Wragg, R. Scotti, I. I. Aguirre, R. Bonsignore, A. Casini, and D. Lovison. "Polypyridyl Ru (II) tamoxifen analogues: Estrogen receptor-targeted dual photodynamic and chemotherapeutic agents". In: *Journal of Inorganic Biochemistry* (2025), p. 113202.

[17] J. A. Stebani, I. Iribarren Aguirre, D. Wragg, A. Gagliardi, and A. Casini. "A Dynamic Evaluation of Cisplatin Encapsulation into [Pd2L4] 4+ Metallacages in Solution". In: *ACS Nano Medicine* (2025).

[16] J. A. Stebani, I. Iribarren Aguirre, G. A. Siddiqui, D. Wragg, A. Gagliardi, and A. Casini. "Computational workflow to unravel the structural dynamics of supramolecular metallacages in solution". In: *Journal of Chemical Theory and Computation* 21.23 (2025), pp. 12278–12288.

[15] M. Cobas-Carreno, A. Esteban-Martos, L. Tomas-Gallardo, I. Iribarren, L. Gonzalez-Palma, A. Rivera-Ramos, J. Elena-Guerra, E. Alarcon-Martin, R. Ruiz, M. J. Bravo, et al. "Drug screening targeting TREM2-TYROBP transmembrane binding". In: *Molecular Medicine* 31.1 (2025), p. 171.

[14] M. Ferrer, I. Iribarren, T. Renningholtz, I. Alkorta, and C. Trujillo. "Computational design for enantioselective CO2 capture: asymmetric frustrated Lewis pairs in epoxide transformations". In: *Beilstein Journal of Organic Chemistry* 20.1 (2024), pp. 2668–2681.

[13] I. Iribarren, M. Ferrer, I. Alkorta, and C. Trujillo. "Computational Insights into Cinchona-Based Phase Transfer Catalysis for Asymmetric Conjugate Cyanation". In: *European Journal of Organic Chemistry* (), e202400329.

[12] I. Iribarren, E. Mates, and C. Trujillo. "Revisiting ion-pair interactions in phase transfer catalysis: from ionic compounds to real catalyst systems". In: *Dalton Transactions* 53.3 (2023), pp. 1322–1335. ISSN: 1477-9226.

[11] I. Iribarren, C. Trujillo, G. Sánchez-Sanz, E. Hénon, J. Elguero, and I. Alkorta. "Influence of Lewis acids on the symmetric SN2 reaction". In: *Theoretical Chemistry Accounts* 142.8 (2023), p. 81.

[10] M. R. Garcia, I. Iribarren, I. Rozas, and C. Trujillo. "Simultaneous Hydrogen Bonds with Different Binding Modes: The Acceptor "Rules" but the Donor "Chooses"". In: *Chemistry–A European Journal* (2023), e202203577.

[9] N. Melnyk, M. R. Garcia, I. Iribarren, and C. Trujillo. "Evolution of design approaches in asymmetric organocatalysis over the last decade". In: *Tetrahedron Chem* (2023), p. 100035.

[8] I. Iribarren and C. Trujillo. "Efficiency and Suitability when Exploring the Conformational Space of Phase-Transfer Catalysts". In: *Journal of Chemical Information and Modeling* 62.22 (2022), pp. 5568–5580.

[7] I. Iribarren, M. R. Garcia, and C. Trujillo. "Catalyst design within asymmetric organocatalysis". In: *Wiley Interdisciplinary Reviews: Computational Molecular Science* 12.6 (2022), e1616.

[6] N. Melnyk, I. Iribarren, E. Mates-Torres, and C. Trujillo. "Theoretical perspectives in organocatalysis". In: *Chemistry–A European Journal* 28.58 (2022), e202201570.

[5] I. Iribarren, G. Sánchez-Sanz, I. Alkorta, J. Elguero, and C. Trujillo. "Evaluation of electron density shifts in noncovalent interactions". In: *The Journal of Physical Chemistry A* 125.22 (2021), pp. 4741–4749.

[4] I. Iribarren, G. Sánchez-Sanz, J. Elguero, I. Alkorta, and C. Trujillo. "Reactivity of Coinage Metal Hydrides for the Production of H2 Molecules". In: *ChemistryOpen* 10.8 (2021), pp. 724–730.

[3] I. Iribarren and C. Trujillo. "Improving phase-transfer catalysis by enhancing non-covalent interactions". In: *Physical Chemistry Chemical Physics* 22.37 (2020), pp. 21015–21021.

[2] I. Iribarren, G. Sánchez-Sanz, and C. Trujillo. "Anion Recognition by Neutral and Cationic Iodotriazole Halogen Bonding Scaffolds". In: *Molecules* 25.4 (2020), p. 798.

[1] I. Iribarren, M. M. Montero-Campillo, I. Alkorta, J. Elguero, and D. Quiñonero. "Cations brought together by hydrogen bonds: The protonated pyridine–boronic acid dimer explained". In: *Physical Chemistry Chemical Physics* 21.10 (2019), pp. 5796–5802.